

Uncovering the necessary hard- and soft-skills to get IT personnel jobs – A text mining approach

Florian J. Meier

Schöller-endowed Chair for Information Systems
Friedrich-Alexander-Universität Erlangen-Nürnberg
Nürnberg, Germany
florian.j.meier@fau.de

Sven Laumer

Schöller-endowed Chair for Information Systems
Friedrich-Alexander-Universität Erlangen-Nürnberg
Nürnberg, Germany
sven.laumer@fau.de

ABSTRACT

With digitalization disrupting organizations every day, the necessary skills to get a job in modern societies are changing rapidly. Individuals and educational institutions seek knowledge of what industry demands in terms of hard- and soft-skills to plan their next step in having these skills in their personal or educational portfolio. To find out which hard- and soft-skills play a key role in finding IT personnel jobs, we employ a text mining approach on job advertisement data and generate a framework for IT personnel jobs. After doing some standard preprocessing steps, we use topic modeling to generate our results and map those with the IT personnel framework. Our results show that three competencies' fields are relevant for IT personnel jobs, and hard- and soft-skills play both a major role in those. Depending on which competencies field one wants to explore, we highlighted the most necessary skills that may play a decisive role when applying for a job in that position. Our findings contribute to the stream of literature equipping both educational institutions and individuals with recommendations of the skills to include and focus on regarding IT personnel.

CCS CONCEPTS

• **Social and professional topics ~ Professional topics ~ Computing profession ~ Computing occupations**

KEYWORDS

Text Mining, IT Personnel Job Skills, Topic Modeling, Natural Language Processing

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1 INTRODUCTION

Since job characteristics in areas related to digitalization are changing rapidly, educational institutions and individuals are seeking knowledge about the most demanded skills. The changes are due to technological novelties including new approaches and methods requiring humans to frequently adapt and overcome technological and personal skill challenges [21, 43]. Higher educational institutions are encouraged to equip students, e.g., in information systems (IS) programs, as future employees, with the necessary skills and capabilities to land jobs in these fast-changing environments. These skill sets compromise subject-specific skills (hard-skills), such as certain computer languages and domain knowledge, as well as personnel and social skills (soft-skills) preparing them for managing digital change challenges [18]. Furthermore, people are constantly on their personal mission of equipping with the right skills. The necessary skills to get a job do not only cover the status of a certain industry snapshot to a specific point in time but also prepare future employees for changes [21].

Knowing which skills and capabilities jobseekers should have to land a job in the area of digitalization is an ongoing task. Requirements for job seekers are frequently changing and people need recommendations on what to look out for to build their individual skilling-paths. Hence, we ask the following research questions:

What are the key demanded hard- and soft-skills by employers for IT personnel jobs?

We scanned existing literature to identify related work and approaches that consider hard- and soft-skill detection methods in different industries to answer the proposed research question. We then crawled 6824 job advertisements on LinkedIn, a professional online social network for organizations that look for new employees and humans that look for new jobs. After conducting data preprocessing steps, we scanned $n=2596$ of these job advertisements to identify frequently mentioned search terms which we later vectorized and calculated to topic maps, structurally grouping and matching them to a beforehand created IT personnel skill framework. We revealed 22 different skill topics in our database, which consists of various hard- and soft-skills for IT personnel jobs. We then discuss our results and

implications for practice and conclude with the limitations of our study and an overview of future research approaches.

2 RELATED WORK

Scanning the stream of literature on Google Scholar and the online databases Scopus, we identified several approaches that regard job skills for occupations from the industry point of view. These approaches cover skill expectations guided by technological change that lead to changing work environments [20, 32]. Most of these studies use empirical approaches, either quantitative methods like surveys or questionnaires, or qualitative ones guided by interviews with industry experts. An example of one early study regarding job skills was conducted by Benbasat et al. [1980], followed by Igbaria et al. [1991] and Leitheiser [1992] using the methodological approach of standardized questionnaires. The respondents of these questionnaires were either employees [14], or, as the majority, employers [16, 25, 34]. Fewer papers used qualitative approaches to identify industry expected job skills by interviewing either employers or managers [11, 17, 33, 41].

Next to these classical quantitative and qualitative approaches to identify job-skills, other methods surrounding manual, semi-automatic or automatic natural language analysis were conducted in the past. These data-driven approaches formerly included manual analysis of newspaper job advertisements [3, 28, 44], which influenced modern data science approaches from the field of natural language processing and text mining. These modern approaches make inter alia use of online social networking sites (such as LinkedIn) to generate large amounts of textual data representing job advertisements. A study by Aken et al. [2010] is an example of such a method, as they extracted nearly 250,000 job advertisements from the IT industry.

The general aim of text mining is discovering knowledge, such as patterns and trends, from large-scale textual data. In a more specific way, this refers to identifying relevant words in a document, word relationships, mood, and semantical information to draw conclusions or categorize data [37]. By analyzing large amounts of textual data, this approach may overcome limitations found in traditional qualitative and quantitative research methods. For example, surveys and questionnaires with closed-ended questions can only measure predefined constructs, while in-depth interviews are resource-intensive and heavily dependent in personal contacts [39].

To generate knowledge about the applicability of natural language processing, and especially text mining, concerning job advertisements in this paper, we conducted a literature review on the Scopus database with keywords including ‘text mining’, ‘natural language processing’, and ‘job advertisement’. This literature search revealed that researchers from different disciplines are concerned with natural language processing and text mining on job advertisements. In the following, we present an excerpt of the identified related work:

To analyze labor market trends, Papoutsoglou et al. [2022] develop a framework using LinkedIn job advertisements and apply it to the case of the electric vehicle industry. Their overall

objective is to detect the skills sought by the electric vehicle industry using different methods of text mining. This study emphasizes the need for a framework or taxonomy to classify the findings of textual mining approaches.

Another study by Binici [2021] aims to determine the information technology skills needed in information institutions e.g., libraries, from job advertisements between 2004 and 2020. The authors identify several hard-skills demanded by the information institution industry, including programming languages, operating systems, and databases. Pejic-Bach et al. [2020] identify skill-sets necessary for industry 4.0 jobs from online job advertisements, defining conceptual solutions to identify needed skills in upcoming and changing industries. Föll & Thiesse [2017] proposed an approach to align information system curricular designs by skills actually needed from industry, especially for job starters and young graduates. The authors use job advertisements and a topic modeling approach to identify and classify their findings. The three papers further support the applicability of textual mining approaches in different industries.

In regards to soft-skills as demanded industry skills, Calanca et al. [2019] studied soft-skill requirements in job advertisements, highlighting their relevance, especially in female-oriented professions. To extract the skills, the authors also make use of text mining methods. Paklina & Shakina [2021] evaluated the importance of skills in digital transformation, aiming at identifying the new composition of skills and their value as implicitly manifested by employers when they look for the new labor force by text mining over 9 million job advertisements from a Russian job board between 2006 and 2018. Others evaluate the automotive market [29], or certain occupational types, such as statisticians [27] or data scientists [2].

Regarding the approach of the text mining methods, all papers employ topic modeling to analyze their findings. However, within topic modeling, the identified studies use different algorithms to utilize the approach. Those include Latent Dirichlet Allocation (LDA) [7], or Seeded LDA [42].

This short literature review on related work regarding this study shows the applicability of text mining to identify demanded skills by industry. It further highlights the importance of public job advertisements as a data source.

We could not identify a recent study that reveals hard- and soft-skills for IT personnel jobs regarding industry needs. Hence, we are addressing this research gap by analyzing job advertisements from LinkedIn, build an IT personnel skill framework and categorize our findings within that framework.

3 CONCEPTUAL APPROACH

This chapter explains our conceptual approach to answering the proposed research question on what hard- and soft-skills are necessary for IT personnel jobs. To do so, we generate a database of job advertisements which we then analyze, after some data cleaning and preprocessing, with topic modeling using LDA [7]. We further establish an IT personnel skill framework to map our topic models to job skill areas and competencies fields.

3.1 Data Collection and Preparation

To answer the proposed question, a thorough database was needed. As we could not identify a publicly available dataset that covers job advertisements for IT personnel, we established a web crawling approach to generate such data. We decided to focus on the professional online social network LinkedIn, as it is among the most popular platforms in publishing job advertisements covering a broad range of companies and organizations with no primary focus on any given industry or job type [9]. According to Collmus et al. [12] 94 percent of hiring professionals use LinkedIn for recruitment processes, and 40 percent of job seekers use the platform to look for suitable professions. To retrieve the necessary data, we developed a web scraper using the Python programming language that mimics a human visiting LinkedIn with the search term “Informatik” – the German equivalent to computer science – to include both technical jobs, but also one’s regard information systems, as the German expression is “Wirtschafts-Informatik”. The web scraper works in two steps: It first generates a list of URLs that link to job advertisements in the respective field and stores them in a .csv data file. It then automatically goes through every URL to scrape off the data for the demanded job skills of every individual job advertisement. This process yielded 6824 job advertisements in June 2021.

To obtain good results, we needed to remove duplicates from our generated database and job advertisements targeted at undergraduate students or apprenticeships. We further excluded demanded skill descriptions with 500 or fewer characters to enhance our results’ reliability, yielding a total of 2596 job advertisements for our analysis.

3.2 Data Analysis – A Topic Modeling Approach

Our database consists of unstructured textual representations of job advertisements that were previously posted on the online social network LinkedIn. To obtain knowledge about the demanded IT personnel job-skills, we relied on text mining methods to further prepare and process our database. These methods generate topic models consisting of semantically coherent words to topic bases.

3.2.1 Textual Preprocessing

We followed the recommendations of Boyd-Graber et al. [2014] and Schmiedel et al. [2019] for preprocessing textual structures, adding four steps to the data analysis: The first step is transforming the whole corpus into lowercase and tokenize the words, which is standard in preprocessing [45]. The transformation is necessary to uniform the corpus and make it better analyzable. The second step is removing words composed of less or more than a specific count of characters. Words that were less than one character were removed from the text corpus, but no maximum word limit was used. An exception was made for ‘R’ and ‘C’, as they both may refer to the programming languages. The third step is removing so-called “stopwords”, which are common words with no specific value for the analysis

[6]. These words include phrases like “the”, or “that”. The fourth step is “stemming” words, which refers to removing suffixes. The general objective of this step is to replace words like “values”, “valued”, and “valuing” with “valu” to identify them as one word. This process yielded a textual corpus that fulfilled the requirements for further textual analysis.

3.2.2 Defining optimal Topic Number

To obtain analyzable results with topic modeling and Blei et al.’s [7] LDA, we need to define the optimal number of topics. Furthermore, two hyperparameters α and β need to be specified. The LDA approach using the Gensim framework [38] allows for automatic estimations of the hyperparameters α and β , but we need to adjust our topic number manually.

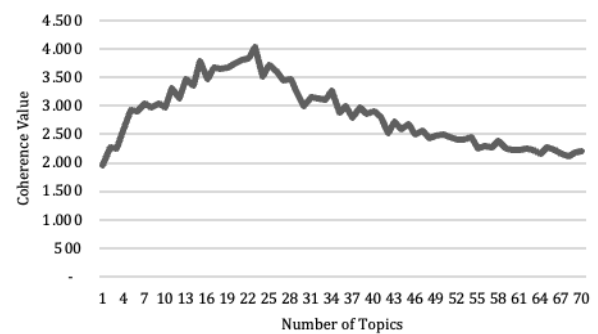


Figure 1: Defining the optimal topic number

Different approaches are available to determine the topic number, such as qualitative analysis by coders [19] or calculating the semantic coherence between topics [31]. We choose to incorporate the latter by calculating a coherence score with the ‘uMass’ method. However, our primary goal was to check that human interpretability remains when choosing our topic number. The results of our coherence calculation approach, which is graphically displayed in Figure 1, revealed that the highest coherence score is around 22 topics, which is an acceptable score to make sure topics are clearly distinguished from each other but remain interpretable. Empirical research has shown that human-interpretable topic models normally include 10-50 topics [39].

3.2.3 Topic Modeling with LDA

To divide our textual corpus into different semantical partially independent topics, we use Blei et al.’s [7] LDA, which randomly divides all words of the corpus into a given number of topics. Each word is checked to correspond with the remaining words in the topic. The procedure assumes that all other words are correctly distributed among the topics. If the word does not match the expected topic, it is assigned to the topic best fitting, otherwise, it remains. Again, we use the Gensim framework [38] to implement the unsupervised algorithm in the Python programming language.

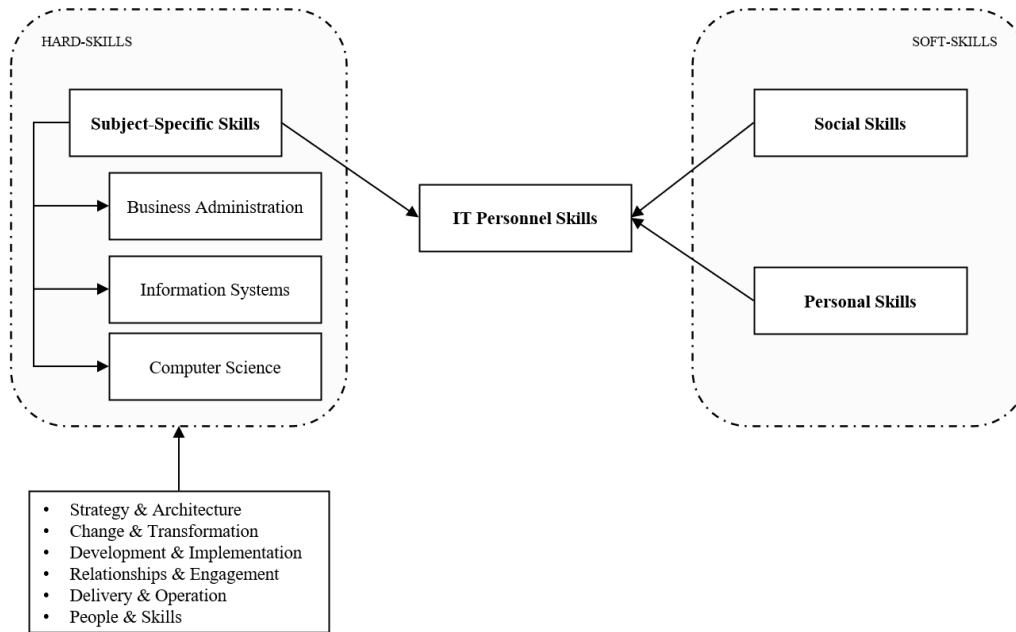


Figure 2: IT Personnel Skill Framework

The individual 22 topics we obtained from this approach were evaluated to check the words in the topic for fitness. We decided to solely check the top seven words of each topic as literature from psychology recommends seven plus/minus two as the “magical” number for humans to process artifacts and information [30]. We manually switched words in 6 topics that seemed misplaced in the top seven with the ones that appeared later in the topic but are still in the top 20.

3.2.4 IT Personnel Skill Framework

To map our calculated topic models, we developed an IT personnel skill framework helping us structure our results. This framework is fundamentally based on (1) the Guidelines for Education in Business and Information Systems Engineering [23] and (2) the Skills Framework for the Information Age [40].

Skills can generally be split in two categories: hard-skills, which are subject-specific, and soft-skills. In this paper, we look at IT personnel hard-skills as a skill-subset from business administration and computer science, supplemented by business and information systems engineering skills [23], which can be seen on the left side of Figure 2. The skills are necessary to understand various domains of IT application within organizations, develop, manage, and understand software and information systems in single instances and between organizations, and exploit various benefits of IT in organizations. According to [40], there are six different fields of competencies within the subject-specific hard-skills that are within or between the hard-skill-subsets. These cover the different skill and job areas of IT personnel, namely ‘Strategy & Architecture’, ‘Change & Transformation’, ‘Development & Implementation’, ‘Relationships & Engagement’, ‘Delivery & Operation’, and ‘People & Skills’. These skill areas further include sub-areas regarding certain job

types, e.g., the development and implementation field of system developers. In this interdisciplinary nature of IT jobs, a clear assignment of the competence fields to the hard-skill subgroups business administration, information systems, or computer science is not possible.

In regard to soft-skills, which can be seen on the right side of Figure 2, we can differentiate between social and personal skills [23]. On one hand social skills are the ability to lead teams, showing interest in other people and their concerns, and include aspiring character traits such as teamwork abilities, cooperativeness, negotiation and discussion skills, and reliability. On the other hand, personal skills concern with analyzing, presenting, communicating, problem-solving abilities, customer orientation, and project management [23]. Figure 2 depicts our IT Personnel Skill Framework which represents the certain skill areas for IT personnel jobs.

4 RESULTS

Our topic modeling approach yielded 22 topic maps which we mapped on the IT personnel skill framework. Six identified topics were assigned to the soft-skill part of the framework, while 16 landed in the hard skill sector. Within the latter, and as expected, we could not find clear distinctions between business administration, information systems, and computer science regarding the certain topics. Hence, we categorized the topics under the competencies field ‘Strategy & Architecture’, ‘Change & Transformation’ and ‘Development & Implementation’. There were no clear topics found for IT personnel skills that belonged to the competencies’ fields of ‘Relationships & Engagement’, ‘Delivery & Operation’ and ‘People & Skills’.

IT Personnel Skills									
Strategy & Architecture		Change & Transformation	Development & Implementation				Social Skills	Personal Skills	MS Office
Strategy & Planning	Security & Privacy		System Development	Data & Analytics	Cloud Development	Content Management			
2 Topics HS: agile_method scrum canban ms_office SS: communication presentation language travel_ready structure customer_orientation independence	1 Topic HS: tcp tcp-ip firewall certificates SS: analytical_thinking independence autonomy	1 Topic HS: business knowledge ms_office SS: motivation trend_awareness responsibility perform flexibility	5 Topics HS: java python .net visual_studio sql sap javascript html css angular php typescript git SS: communication teamwork	4 Topics HS: python matlab_simulink r sql java mathematics statistic SS: independence self_initiative	1 Topic HS: docker kubernetes azure scrum kanban aws java	1 Topic HS: jira_confluence java spring git scrum SS: agility	3 Topics SS: communication team_orientation on language reliability	3 Topics SS: structured_working customer_readiness readiness_development presentation analytical_thinking	1 Topic HS: Ms_office Excel Sql mathematics SS: Operational Independence Team_orientation

HS = Hard Skill
 SS = Soft Skill

Figure 3: Topics in the IT Personnel Skill Framework

We assigned three topics to our first skill field, ‘Strategy & Architecture’. We further subdivided the field into two topics belonging to ‘Strategy & Planning’ and one to ‘Security & Privacy’. The latter could also be assigned to ‘Development & Implementation’. The two topics in ‘Strategy & Planning’ compromise task-fields that include the sustainable strategic alignment of organizations, covering the hard-skills using agile methods, such as Scrum, Kanban, and Microsoft Office. Especially the latter is apparent through multiple topics and an important factor in the IT personnel job hard-skill set. Another factor that protruded in this category was the strong occurrence of soft-skills. These include communication skills, presentation skills, language skills (esp. English), travel readiness, structuredness, customer orientation, and independence. The one topic regarding ‘Security & Privacy’ includes hard-skills around the protocol TCP-IP, firewall-knowledge, security certificates, and analytic conceptual development. This topic gets fulfillment by analytical thinking skills, independence, and autonomy. Jobs in this field cover strategy and planning professionals who manage the next organizational steps, assess risks of both technical and economical nature and advise other employees [40].

Our second respective hard- and soft-skill field is ‘Change & Transformation’, where soft-skills showed a significantly higher presence than their hard-skill counterparts. We assigned one topic to this field consisting of the hard-skills business knowledge and MS office, and several soft-skills enabling leadership and change. These soft-skills are motivation, trend-awareness, responsibility, willingness to perform, and flexibility. The said skill set is expected to prepare for changing environments, frequently adapting to new challenges and leading organizational digital change challenges due to lowering perceived technology impacts

on employees and work routines through enhanced flexibility [24]. Jobs in this regard include portfolio and program managers, change analysis, and change planners [40].

The biggest competencies field is ‘Development & Implementation’, comprising 11 topics. In this field category, five topics are assigned to ‘System Development’ covering the necessary hard- and soft-skills for this job. The most prominent hard-skills are Java and Python, the .NET framework, the visual studio IDE, SQL, SAP, web development skills covering JavaScript, HTML, CSS, AngularJS, php, TypeScript, and the version control software git.

Soft-skills took only a minor role in this category. The present soft-skills include communication skills and the ability to work as a team. Another big topic in the ‘Strategy & Development’ category is ‘Data & Analytics’ with four topics present. Here, hard-skills like Python, Matlab Simulink, R, database skills such as SQL and overall relational database skills, Java, Math- and statistical skills are present. Again, soft-skills play a minor role in this area, with independence, self-initiative, solution- and team-orientation being present. Another topic in this field regards ‘Cloud Development’. We found the hard-skills Docker and Kubernetes, Microsoft Azure, Scrum, Kanban, AWS, and Java present in this topic. No soft-skills were found in this topic, which does not exclude their occurrence in real-world settings. The ‘Development & Implementation’ competence field concludes with ‘Content Management’ competencies, which include the hard-skills Jira Confluence, Java, the Spring framework, git, and Scrum. Furthermore, the soft-skill agility was present in this topic. The present jobs in this field are manifold, but foremost of technical nature. One big job part regards the development of new systems, both in hard- and software, and includes testing and

configuration. Furthermore, data analyst jobs are on the rise [13], which is also present in our findings. Tasks for this job type include data management, engineering, analysis, and visualization, database design and administration.

One discovered topic fell out of the line of our IT personnel skill framework, covering solely Microsoft Office-related skills. Of course, the hard-skills present are the MS Office package and especially highlighted Excel, mathematics, and SQL. This was fulfilled with soft-skills of operational readiness, independence, and team-orientation. We expect MS Office competencies to be a core skill for IT personnel, nearly regardless of the job.

For the soft-skill part of our IT personnel framework, we identified six topics with three in social and three in personal that happened to include fitting content. The most present skills belonging to social skills are communication skills, team-orientation, language skills, that foremost include the English language, and reliability. Personal skills are structured working, readiness for further development, customer orientation, presentation skills, and analytical thinking. While these cover the most prominent findings from the soft-skill topics, we want to advise that, like in other fields, e.g., finance [15], soft-skills play a heavy role in getting an IT personnel skill job. One should not neglect the crucialness of soft-skills on their journey to complete their skill sets. We graphically display our findings in Figure 3, including the identified competence areas and various hard- and soft-skills.

5 DISCUSSION AND IMPLICATIONS

Our research was motivated to detect the key hard- and soft-skills for IT personnel jobs. Reviewing the 22 topic maps in our results, we could identify that the key hard-skills belong to three areas: ‘Strategy & Architecture’, ‘Change & Transformation’ and ‘Development & Implementation’, focusing on business tasks and digital change or developing new software. Furthermore, soft-skills are present in all areas of IT personnel jobs. Individuals who are on their path to getting into the IT industry should consider mixing up their hard- and soft-skill resumes, focusing on one of the three key skill areas. We could identify a strong occurrence of soft-skills in almost every topic model of our results. This finding further indicates the prevalence of soft-skills in the IT industry.

We will discuss the implications of our research in more detail in the following:

First, given the review in section two on related work and the different approaches of identifying demanded skills of professionals from the past and the present, we contribute to the stream of literature by providing a text mining approach of job advertisements. Previous work identified different demanded hard- and soft-skills for certain industries [29], jobs [13], or university curricular [19]. We contribute by starting a new discussion about hard- and soft-skills for IT personnel jobs, breaking them down to the lowest granularity of certain skills within skill topics. Our conceptual approach, including the data analytics steps and the IT personnel framework skill framework, can be reproduced in other industries to identify the necessary hard- and soft-skills present there.

Second, considering the general call of research to investigate university curricular design, we highlight relevant skills necessary to equip future employees for their jobs. Educational institutions should reflect whether they align with the demands of the economy and society, to follow the fast-changing nature of IT-related jobs [19]. This is valid for both hard- and soft-skills. For example, institutions could reflect on whether the used programming languages need to be replaced by ones used in industry for certain tasks (e.g., Python), university modules could focus on knowledge management frameworks that are common to preserve knowledge in organizations (e.g., Jira Confluence), or could employ mandatory parts in their teaching to foster soft-skills and team-work. We plan on expanding our findings on a broader level, comparing the occurrence of research findings of hard- and soft-skills with the ones taught in educational institutions to reveal variances and propose for change. This also calls for a future design-oriented approach, creating an IT artifact that automatically not only reveals shortcomings in curricular design but recommends suitable actions and curricular design recommendations. On an individual level, such a design-oriented approach could recommend suitable next steps in fulfilling skill-sets with an overall job objective based on real industry job advertisement data. This would help individuals choose suitable modules and trainings to reach their next career step.

Nonetheless, our approach and contributions are limited by some facts. First, we only consider a German data sample, limiting the results geographically. To further extend our results, a geographically wider data set is necessary to reveal differences among countries and cultures. Second, we only used a data snapshot from one point in time for our study. To reveal further findings and enhance the reliability of our results, a longitudinal study is necessary that confirms or falsifies the presence of certain skills and skill topics in the industry. Furthermore, we only used Blei et al.’s [7] LDA for our analysis. Hence, different textual data analytic approaches could be employed that may generate a new perspective. Also, supervised or semi-supervised approaches could be respected for the data analysis to get another view on textual data.

Given these limitations and considering the discussed implications future research can use our approach to further extend studies on hard- and soft-skills based on industry needs. First, future research can replicate our data analysis with new data and from potentially geographically different origins that may shed light on the skills for IT personnel jobs. As already highlighted, future research could also employ design-oriented research building IT artifacts that guide educational institutions or individuals. Furthermore, a thorough literature review is needed that reveals the differences among the various studies of (semi-) automatic textual analysis with job advertisement data. Such a detailed review is expected to reveal different data analytic approaches with various algorithms that may produce interesting results. Especially regarding other disciplines, such as computer science, a literature review is expected to reveal new opportunities for textual data analysis with job advertisements.

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